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# Exploring relationships between individual differences and learning outcomes in data-driven learning: A meta-analytic path analysis approach

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## Abstract

Although the pedagogical benefits of data-driven learning (DDL) are well established in language learning, the role of individual differences—such as learners' demographic, cognitive, and affective factors—has received limited systematic attention. To address this gap, this study employed meta-analytic structural equation modeling to identify individual difference variables and analyze the structural relationships influencing language learning outcomes in DDL. Based on 25 studies ($N$ = 1094), this analysis identified the following key predictors of DDL effectiveness: learners' working memory, L2 proficiency, positive perceptions of DDL, and familiarity with DDL (including corpus-consultation skills). Structural relationships revealed that L2 proficiency (β = 0.20, SE = 0.07, $p$ = .003) and familiarity with DDL (β = 0.06, SE = 0.03, $p$ = .046) significantly impacted learning outcomes. Additional analyses explored potential moderating factors, such as target-language domains, L1-L2 relatedness, instructional mode, and a concordancing approach; however, no significant moderating effects were observed.

**Keywords:** Individual differences; Data-driven learning (DDL); Meta-analytic path analysis (MAPA); L2 proficiency; Familiarity with DDL

## 1. Introduction

The pedagogical benefits of data-driven learning (DDL) are well recognized in the language learning field, with a significant number of primary studies having demonstrated its effectiveness. Regarding how the effectiveness of DDL varies according to individual differences, recent meta-analyses (Boulton & Cobb, 2017; Lee, Warschauer, & Lee, 2019; Ngo & Chen, 2024) have identified learners' first language (L1), second language (L2) proficiency, discipline, and educational level as significant moderators that account for language learning through the DDL approach. However, some researchers (Chambers, 2005; Lee, Warschauer, & Lee, 2020; Yoon, 2008) have argued that apart from these moderators, learners' cognitive and affective factors may play a crucial role in determining the pedagogical benefits of DDL, although the field has seen little synthetic effort to examine the roles of these factors. That is, although the aforementioned meta-analyses (Boulton & Cobb, 2017; Lee et al., 2019; Ngo & Chen, 2024) have successfully demonstrated the effectiveness of DDL over a business-as-usual approach in terms of L2 learners' language acquisition and identified a range of moderator variables, their syntheses concerned between-group studies (Plonsky & Oswald, 2014), which compare treatment and comparison conditions. To date, no synthetic effort on DDL has considered correlational studies that explore the relationships among key factors contributing to DDL outcomes, despite previous studies (e.g., Lee et al., 2020; Yoon, 2008) highlighting the importance of these factors.

Therefore, the field critically needs correlational studies, particularly those employing structural equation modeling (SEM) in varied learning contexts, to address current research gaps. For instance, Lee et al. (2020) used SEM to explore how learner factors like prior English proficiency, working memory, and DDL strategy use shape DDL achievement. Despite this important contribution, there's a scarcity of follow-up SEM studies to validate and extend their findings. Recognizing this limitation, we sought a method that could go beyond single-study replication. Our solution was to employ a meta-analytic structural equation modeling (MASEM) approach, which enabled us to aggregate and analyze correlation matrices from multiple studies. It thus provided a more comprehensive and statistically robust means to validate and extend the existing SEM findings by incorporating additional individual learner difference factors that were not addressed in previous SEM models, and to further advance the understanding of DDL mechanisms.

As its name suggests, MASEM combines meta-analysis (MA) and structural equation modeling (SEM). It involves collecting bivariate correlation coefficients from the literature and synthesizing them into a pooled correlation matrix using weighted calculations based on sample sizes. This pooled matrix is then used to assess whether it aligns with the hypothesized

structural model. Additionally, MASEM allows us to examine whether moderators significantly influence the structural relationships among the variables in the model, specifically focusing on the path coefficients that link them. In doing so, our goal was to identify individual difference variables and investigate the complex relationships between these variables and language learning outcomes in DDL. By employing these statistical techniques, we aim not only to examine the relationships between individual factors but also to understand, from a holistic perspective, how these variables are organically interconnected and collectively influence DDL outcomes.

## 2. Pedagogical benefits and practical challenges of DDL

DDL refers to an approach in second language acquisition (SLA) that utilizes large structured language datasets, also known as linguistic corpora, to facilitate language learning (Johns, 1994). The core idea is that learners engage with authentic linguistic data to uncover patterns, structures, and usage in context, rather than relying solely on prescriptive rules or explicit instructions. Research into DDL has identified several pedagogical benefits for language learners, although it has also highlighted practical challenges that must be considered to make this approach effective.

Among the primary benefits of DDL is its emphasis on authentic language use. By working directly with real-world language examples—whether from texts, transcriptions, or spoken corpora—learners are exposed to linguistic variability and contextual nuances that are often absent in textbook materials (Boulton, 2010). This exposure helps learners develop a more nuanced understanding of a language, especially in terms of grammar, collocations, and idiomatic expressions.

Moreover, DDL supports learner autonomy by affording opportunities for self-directed learning (i.e., discovery learning; Flowerdew, 2015). Learners can explore corpora on their own, navigating data to identify language features that interest them, such as collocational patterns (Boulton & Cobb, 2017). This process encourages meta-cognition as learners develop skills to independently analyze and interpret language. According to Boulton (2017), this form of learning aligns with constructivist principles, whereby learners actively build their knowledge through interaction with language rather than passively receive information.

Furthermore, DDL helps integrate form and meaning, which is critical for L2 learning (VanPatten, Williams, Rott, & Overstreet, 2004). By analyzing data from various contexts, learners acquire not only how language forms are used but also how they convey meaning in specific communicative situations. For instance, a learner might observe through corpus data

how different verb forms are used in formal versus informal contexts, or how certain lexical choices are associated with specific situations (e.g., cooperate or coordinate; Lee, Shin, & Lee, 2012).

Despite its advantages, implementing DDL in classrooms entails several practical challenges. A major issue concerns the technical proficiency needed to use corpus tools and analyze large datasets effectively. Although more user-friendly DDL resources are now available (Lee et al., 2015; McEnery, Xiao, & Tono, 2006), many teachers and learners still lack the skills required to employ them efficiently, especially those without training in computational linguistics or corpus-based methods. For learners, particularly weaker ones, the process can be overwhelming (Meunier, 2020), as they may struggle to interpret concordance lines or connect large amounts of language data to learning tasks. While DDL promotes learner autonomy, it also requires scaffolding to ensure accurate interpretation and application of corpus insights. Without such support, learners risk misinterpreting or overgeneralizing patterns, potentially leading to errors (Lee et al., 2017). Hence, teachers are encouraged to provide strategic guidance to help less experienced learners navigate DDL tasks and develop the ability to draw meaningful language insights (O'Keeffe, 2021).

Another critical challenge lies in the quality and relevance of the corpora available for classroom use. Teachers and learners often struggle to find datasets that match their L2 proficiency levels, specific learning goals, or personal interests (Flowerdew, 2012). General-purpose corpora such as the British National Corpus (BNC Consortium, 2007) provide broad language coverage but may include content that is too complex or domain-specific for many learners (Lee et al., 2017, 2019). As a result, teachers must often curate or adapt materials to reflect their students' needs, which demands considerable time and effort.

## 3. Role of individual differences in DDL

In this section, we introduce the cognitive and affective factors that have been regarded as key variables of individual learner differences in the context of DDL, along with those expected to moderate the relationships among them.

### 3.1. Cognitive factors

Cognitive factors significantly influence how learners interact with and benefit from DDL. Since data-driven learning relies on learners' active engagement with authentic language data, individual differences in working memory, L2 proficiency, prior experience with DDL tools (i.e., DDL familiarity), and DDL strategy use can substantially shape how effectively they process corpus information and extract meaningful patterns.

### *3.1.1. Working memory*

Working memory, a key cognitive system that enables the temporary storage and manipulation of information during learning tasks (Baddeley, 2003; Linck, Osthus, Koeth, & Bunting, 2014), has been identified as a crucial factor influencing DDL performance. Learners with stronger working memory are better equipped to handle the cognitive demands of navigating corpus tools and interpreting concordance lines (Lee et al., 2020). Conversely, learners with limited working memory may feel overwhelmed by the volume of data presented and struggle to extract meaningful patterns. This disparity underscores the need for instructional strategies that reduce cognitive load, such as breaking tasks into smaller steps or using visual aids to guide learners through the process (Gathercole & Alloway, 2007).

### *3.1.2. L2 proficiency*

L2 proficiency is another critical factor that shapes learners' ability to engage with DDL (Boulton & Cobb, 2017; Lee et al., 2020). Beginners may find it challenging to interpret the complex linguistic structures and advanced vocabulary often present in general corpora, which can limit their ability to derive meaningful insights (Flowerdew, 2012). By contrast, advanced learners may benefit from exploring nuanced patterns and subtleties within corpora, as these align better with their linguistic competence (Breyer, 2009). Thus, L2 proficiency determines the extent to which learners can effectively process corpus data and transform them into meaningful linguistic knowledge.

### *3.1.3. DDL familiarity*

Furthermore, despite the growing availability and accessibility of corpora and tools (e.g., Lee et al., 2015), prior experience with DDL tools significantly influences a learner's ability to effectively navigate corpora. Learners who have attended training sessions or previously used corpus tools are more likely to have developed robust corpus-consultation skills, which are recognized as effective strategies in DDL (Lee et al., 2020). These skills enable them to formulate more precise queries, accurately interpret concordance lines, and extract relevant data with greater ease (Boulton, 2010). By contrast, learners with limited exposure to these tools may require additional support and guidance to build their proficiency and confidence in using them for language learning tasks (Lee et al., 2019).

### *3.1.4. DDL strategy use*

Strategy use could be another key determinant of learners' success in DDL. Studies on lexical inferencing (e.g., Anvari & Farvardin, 2016; Hermagustiana, 2018; Hu & Nassaji, 2014) indicate that the quality of learners' strategy use and their strategic deployment significantly influence lexical inferencing. Although the aforementioned studies do not focus specifically on

DDL, given that lexical inferencing constitutes a core component of DDL (Flowerdew, 2025), it can be inferred that effective strategy use is closely tied to successful DDL engagement. Lee et al. (2020) supported this view by identifying the strategies learners employed in DDL-based vocabulary learning and demonstrating that the degree of strategy use had a substantial impact on vocabulary learning outcomes via DDL. Therefore, understanding how learners employ and refine these strategies may offer valuable insights for enhancing the effectiveness of DDL-based instruction.

### 3.2. Affective factors

In addition to cognitive factors, learners' affective factors are equally crucial in determining how they engage with DDL. Variables such as motivation, anxiety, enjoyment, and learners' perception of DDL can shape their willingness to explore corpora and persist through challenges, ultimately influencing the effectiveness of data-driven learning.

#### *3.2.1. Motivation*

Motivation is a central driver as it influences learners' willingness to explore and persist in self-directed DDL tasks. Highly motivated learners are more likely to actively engage with language learning tools, experiment with queries, and draw meaningful conclusions from the data (Ushioda, 2011). By contrast, learners with low motivation for language learning may perceive DDL activities as tedious or intimidating, particularly if the tasks are not clearly aligned with their personal goals or interests (Boulton & Cobb, 2017).

#### *3.2.2. Enjoyment and anxiety*

Enjoyment and anxiety represent key affective factors that may shape learners' engagement with and outcomes from DDL. When learners view DDL as an enjoyable and empowering approach, they may be more likely to persist in using corpus tools and achieve better learning outcomes (e.g., Spivey, 2023). In contrast, anxiety can act as a substantial barrier to effective DDL engagement. The technical nature of corpus tools and complexity of concordance lines may evoke feelings of inadequacy or fear of failure, especially among learners who lack confidence in their language skills (Horwitz, 2001). Consequently, learners experiencing anxiety may avoid using corpus tools altogether or struggle to use them meaningfully.

#### *3.2.3. Perception of DDL*

Learners' positive perception of DDL also plays a crucial role in determining how effectively they engage with corpus tools. When learners view DDL as a valuable and effective approach to language learning, they are more likely to embrace it with enthusiasm and a sense of agency. A positive attitude toward DDL can enhance learners' confidence in their ability to

make meaningful discoveries from corpus data, leading to a greater willingness to engage in independent learning tasks (Boulton & Cobb, 2017). Additionally, when learners perceive DDL as an effective tool for improving their language skills, they are more likely to keep using corpus tools and integrate them into their learning routines. This perception may be influenced by factors such as the perceived relevance of the data, practical applications of corpus findings, or the ability to track progress through observable improvements in language proficiency (Yang, Chen, & Tian, 2024).

### 3.3. Moderating factors: methodological differences

In addition to the aforementioned factors, a review of prior meta-analyses focusing on between-group studies of DDL (e.g., Boulton & Cobb, 2017; Lee et al., 2019; Ngo & Chen, 2024), together with the methodological differences observed across existing correlational studies, revealed a number of variables that may moderate the structural relationships among the aforementioned cognitive and affective factors. These potential moderators are discussed below.

**First**, studies differed in terms of the *target linguistic dimensions* under investigation. That is, some examined receptive aspects (e.g., receptive knowledge of collocation in Kontsioti, 2024), while others focused on productive outcomes (e.g., writing performance in Vyatkina, 2016). Given the nature of the DDL approach, which provides learners with repeated exposure to target linguistic items (Johns, 1994), Lee et al. (2019) proposed that DDL could foster both receptive and productive L2 learning, albeit to varying degrees. Accordingly, *target linguistic dimension* was included as one potential moderator in the present analysis.

**Second**, the *learning environment* may also shape the relationships among the variables mentioned above. While the majority of DDL studies (e.g., Curado-Fuentes, 2024; Hirata, 2019) have been conducted in classroom contexts, largely because of the method's technical demands and need for instructional scaffolding (see *Pedagogical Benefits and Practical Challenges of DDL* section), recent works (e.g., Zare & Aqajani Delavar, 2023) have explored DDL use in remote or home-based settings. The absence of a teacher's physical presence in such environments may affect the interplay of cognitive and affective factors, and thus *learning environment* was treated as another potential moderator.

**Third**, variations were also found in learners' *L1 backgrounds* (e.g., Thai, Korean, Japanese, Persian), which differ in proximity to the target language (i.e., English) (Beaufils & Tomin, 2020). Both Boulton and Cobb (2017) and Ngo and Chen (2024) highlighted this variability, noting that learners' L1 may influence DDL effectiveness. Given this pattern and the precedent set by prior meta-analyses, *L1–English proximity* was incorporated as a further moderator,

given than structural similarities can facilitate positive transfer (Lee, Jung, & Lee, 2022; Ringbom, 2007) and reduce cognitive load during corpus-based tasks (Sweller, 1994), thereby enhancing learners' capacity to notice and internalize target language patterns.

**Fourth**, *learner age* represents another relevant variable associated with demographic factors. Because DDL in its standard form demands a degree of cognitive maturity and learner autonomy due to its discovery-oriented nature (Flowerdew, 2015), it may be better suited to older learners. Supporting this view, Spivey (2023) argued that socio-culturally mediated and scaffolded forms of DDL may be more appropriate for younger learners. In light of these findings, *learners' age* was also included as a moderator in the current study.

**Finally**, *type of DDL instruction*—that is, *hands-on* (learners directly using concordancing software) versus *hands-off* (teachers pre-process and present concordance data to their learners)—was examined as a potential moderator. Boulton (2012) and Saeedakhtar, Bagerin, and Abdi (2020) emphasized the pedagogical significance of this distinction, suggesting that hands-off DDL mitigates technological barriers as one of the primary challenges in DDL, and may therefore be more suitable for younger or less proficient learners.

## 4. Present study

As mentioned in Section 1, the present study aims to identify individual difference variables and explore the complex relationships between these variables and language learning outcomes in DDL by employing a MASEM approach. The rationale for using this approach lies in the limited empirical efforts to understand the mechanisms underlying successful DDL in language learning, despite numerous studies investigating bivariate relationships between factors and DDL achievement (Boulton & Cobb, 2017). Although the research community broadly agrees that understanding these mechanisms is crucial (e.g., Boulton, 2011), how individual differences, such as cognitive and affective factors, interact to influence the effectiveness of DDL remains unclear. Gaining insights into these relationships would facilitate differentiated instruction, provide tailored support, and allow for a more flexible use of corpus resources, all of which would enhance the effectiveness of DDL and ensure its benefits for a diverse range of learners. To this end, the present study adopts and adapts the conceptual framework proposed by Lee et al. (2020) (see Fig. 1) as a theoretical starting point for conducting MASEM. This model provides a foundation for verifying and refining the structural relationships among learner factors, DDL strategy use, and vocabulary learning outcomes, and will be iteratively refined to incorporate additional individual difference variables should they emerge as empirically or theoretically significant.

To explore such possibilities, the present study was conducted in two phases. In the first phase, we systematically identified and categorized individual difference variables—extending beyond those incorporated in Lee et al.'s (2020) model and including factors that have been reported to function as moderators—across empirical DDL studies through a comprehensive coding procedure. Building upon this foundation, the study aimed to examine (1) how these individual difference variables are interrelated in contributing to the success of DDL activities, and (2) how certain of these variables function as moderators that influence the structural relationships among them. Accordingly, the following research questions guided the analysis:

**RQ1:** How are individual difference variables interrelated in contributing to the success of DDL activities, as examined through a meta-analytic path analysis (MAPA)?

**RQ2:** How do the moderating factors influence the structural relationships among the identified individual difference variables and learning outcomes?

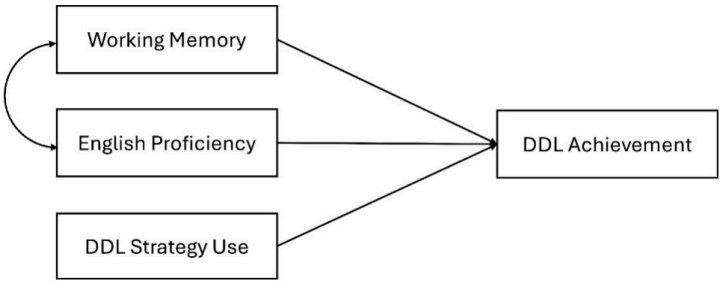



**Fig. 1.** Conceptual framework adapted from Lee et al. (2020) as a starting point for the present study.

## 5. Method

### 5.1. Literature search

To collect relevant studies for the current MASEM study, we systematically searched for empirical studies focusing on DDL in L2 learning published up to the end of 2024. In particular, the search aimed to identify studies that provided correlation coefficients between pairs of variables, including those between individual difference variables and learning outcomes. To ensure comprehensive coverage of language-related research and interdisciplinary studies, five databases were chosen and consulted: EBSCOhost, Linguistics and Language Behavior Abstracts (LLBA), ProQuest (Social Science Premium Collection and Dissertation & Theses Global), Scopus, and Web of Science. A combination of keywords was employed for abstract searches using terms such as [("DDL" OR "Data-Driven Learning") AND "Language Learning" AND ("Individual Difference" OR "Correlation" OR "Factor" OR "Achievement" OR "Outcome")]. The final component of the keyword combination was specifically designed to capture a wide range of studies with quantifiable variables, extending

beyond the initially hypothesized variables to include additional potential, thereby ensuring a broad and inclusive dataset. Five top journals relevant to DDL were consulted: *Language Learning & Technology*, *Computer Assisted Language Learning*, *ReCALL*, *System*, and the *CALICO Journal*. The search also included manual screening of reference lists from relevant articles.

After merging the identified studies and removing duplicates identified from our database, journal, and forward and backward searches, we had a total of 527 studies. We then reviewed their titles and abstracts to assess their relevance to our research topic. Of these, 390 studies were unrelated to DDL, leaving 137 studies for a full review. During the full review phase, we assessed whether the identified studies met our inclusion criteria. Specifically, studies had to (1) be written in English; (2) focus on the use of DDL in L2 learning to enhance learners' L2 achievement; (3) address individual difference variables and learning outcomes; and (4) include at least two quantifiable variables suitable for calculating a correlation coefficient. Following this review, 83 studies were identified as finalists, while 54 studies were excluded for the following reasons: (1) 11 studies were not written in English; (2) 13 were either review articles or conceptual papers; and (3) 30 included only one variable, making it impossible to compute correlation coefficients.

Among the 83 finalist studies, only 21 reported correlation coefficients. To bridge this gap, we emailed the authors of the remaining 62 studies. We explained the objectives of our study and requested correlation coefficients for the variables used in their research. If the authors could not or were disinclined to compute the coefficients, we asked for anonymized raw datasets to calculate them ourselves. To maximize responses, we sent two follow-up reminder emails at two-week intervals and concluded our data-collection efforts after a month. Consequently, we obtained correlation coefficients from nine studies, bringing our final dataset to 30 studies. The literature-search process is summarized in Fig. 2, which follows the PRISMA flowchart format.

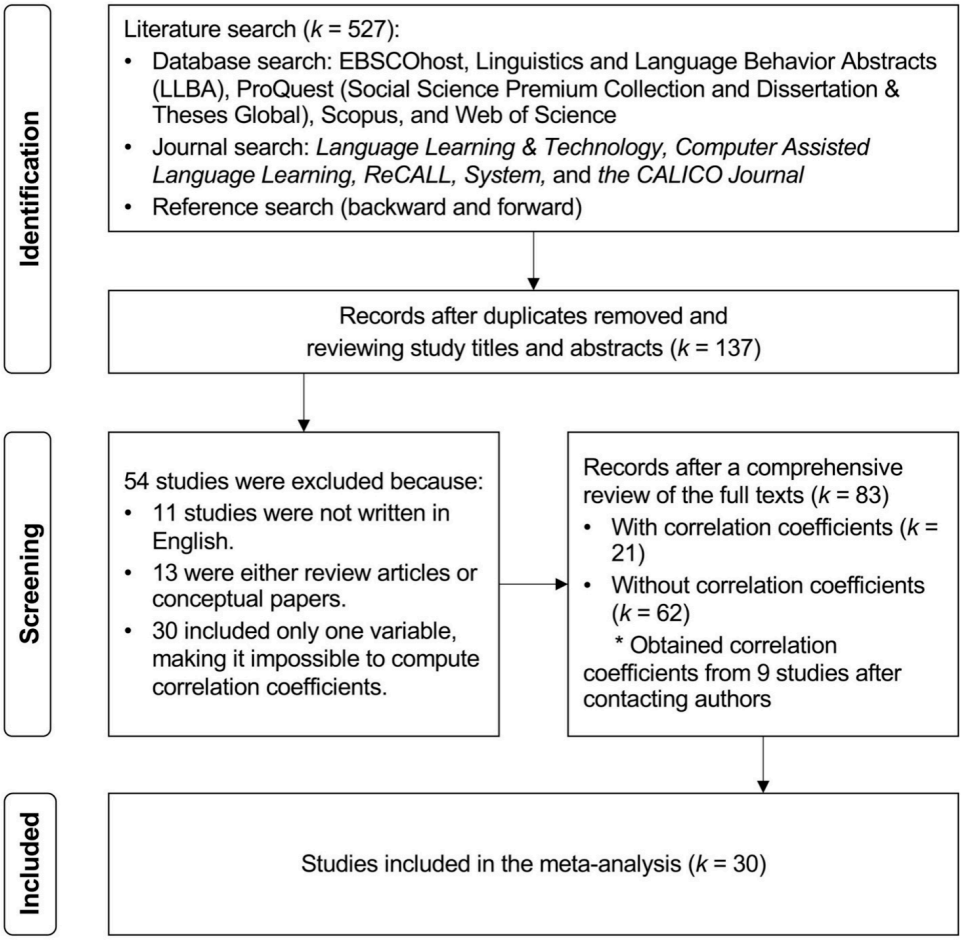



**Fig. 2.** PRISMA flowchart for literature-search process.

## 5.2. Dataset construction

### *5.2.1. Constructing the dataset*

Following the literature search, we reviewed the identified studies to compile a list of the variables investigated. Based on this list, we developed a comprehensive correlation template to extract and organize all correlation coefficients reported in or computed for the identified studies. All coding was performed jointly by the first and last authors, and any disagreements were resolved by discussion to reach a consensus. An Excel file was created, with each study assigned its own worksheet, resulting in a number of worksheets populated with the correlation coefficients collected from the identified studies.

From the 30 collected studies, we identified and coded seven variables: (1) DDL achievement (i.e., learning outcomes); (2) DDL familiarity (e.g., prior experience with DDL, training in corpus consultation skills and strategies); (3) positive perceptions of DDL; (4) perceived effectiveness of DDL; (5) L2 proficiency; (6) L2 motivation; and (7) working memory. These

factors were all drawn from the individual difference variables reviewed earlier (see Section 3, *Role of Individual Differences in DDL*), representing the specific variables that were empirically examined in the studies included in our dataset.

### *5.2.2. Finalizing the dataset*

To ensure that the seven identified variables could be retained in the dataset, we examined whether there was any missingness among the correlations between these variables. Unfortunately, we found that the initial dataset lacked raw correlation data for the following three pairs: (1) between DDL achievement and positive perceptions of DDL, (2) between DDL familiarity and L2 motivation, and (3) between L2 proficiency and L2 motivation. To address this limitation, we made two decisions. First, we removed L2 motivation (Variable 6), as it was involved in two of the three missing correlations. Second, regarding the missing correlation between DDL achievement (Variable 1) and positive perceptions of DDL (Variable 3), we decided—rather than removing Variable 3—to merge positive perceptions of DDL (Variable 3) with perceived effectiveness of DDL (Variable 4) as a last resort, given that Variables 1 and 3 were otherwise well supported in the dataset. This decision was informed by prior research suggesting that perceived effectiveness of DDL (Variable 4) is often treated as a subcomponent of positive perceptions of DDL (Variable 3) in questionnaires (e.g., Liu & Gablasova, 2025; Yoon & Hirvela, 2004). Furthermore, learners' responses to items measuring these two constructs frequently show high similarity and strong correlations (e.g., Alcaraz-Mármol & Soto-Almela, 2022; Kogan, Komarova, & Dmitrijev, 2020).

Following these adjustments, we dropped five studies from the initial dataset and retained 25 studies ($N$ = 1094). The revised dataset thus included five variables: (1) DDL achievement (i.e., learning outcomes); (2) DDL familiarity (e.g., prior experience with DDL and training in corpus consultation skills and strategies); (3) positive perceptions of DDL; (4) L2 proficiency (English as the L2 in all 25 studies); and (5) working memory. Table 1 provides details on the number of raw correlation coefficients and sample sizes of the supporting studies for each correlation included in the pooled correlation matrix. The 25 studies included are listed in Appendix 1, along with their publication types (e.g., journal article, conference presentation, book chapter, or dissertation), and a detailed description of the constructed dataset is provided in Appendix 2.

**Table 1**
Number of raw correlation coefficients and sample sizes for correlations.

| No. | Variable | 1 | 2 | 3 | 4 | 5 |
|---|---|---|---|---|---|---|
| 1 | DDL Achievement | – | 181 | 59 | 523 | 117 |
| 2 | DDL Familiarity | 4 | – | 94 | 34 | 34 |

| No. | Variable | 1 | 2 | 3 | 4 | 5 |
|---|---|---|---|---|---|---|
| 3 | Perception of DDL | 3 | 2 | – | 115 | 192 |
| 4 | English Proficiency | 12 | 1 | 4 | – | 34 |
| 5 | Working Memory | 4 | 1 | 2 | 1 | – |

*Note*. The values below the diagonal show the number of studies that provided correlation coefficients for each specific pair of target variables, while the values above the diagonal reflect the total sample sizes from those studies. Altogether, the dataset comprises 25 studies with a combined sample size of 1094 for the collected correlation coefficients.

### *5.2.3. Revisiting the conceptual framework*

As noted earlier (see Fig. 1), the initial conceptual framework included working memory, English proficiency, and DDL strategy use as predictors of DDL achievement. Based on the dataset constructed for this study and the theoretical considerations described above, we updated this framework to include five variables: DDL achievement, DDL familiarity, perception of DDL, English proficiency, and working memory (Fig. 3). In this revised model, DDL familiarity replaces the original DDL strategy use construct, as it conceptually represents learners' strategic engagement with DDL. In addition, perception of DDL was introduced as a new variable to capture learners' affective and evaluative stance toward DDL use.

The revised framework thus models how English proficiency and DDL familiarity are hypothesized to jointly influence learners' perception of DDL, which in turn predicts DDL achievement (Lee & Lee, 2023). Specifically, greater familiarity with DDL practices is expected to foster more positive perceptions (e.g., Boulton, 2010; Chen, Flowerdew, & Anthony, 2019; Hirata, 2019). Furthermore, learners with higher L2 proficiency are more likely to perceive DDL as a useful and manageable tool for language learning (Boulton & Cobb, 2017; Lee et al., 2020) and to find corpus use less linguistically burdensome, making them more receptive to DDL practices (e.g., Quinn, 2018; Vyatkina, 2016).

Consequently, we proposed the path analysis model illustrated in Fig. 3. In this model, learners' working memory and English proficiency—which are assumed to be correlated—as well as their DDL familiarity, are hypothesized to directly influence their DDL achievement. Additionally, the effects of English proficiency and DDL familiarity on learning outcomes are modeled as being mediated by learners' perception of DDL, which itself also exerts a direct effect on DDL achievement. This updated structure was tested using meta-analytic path analysis (MAPA) rather than full meta-analytic structural equation modeling (MASEM), because the dataset contained only observed variables and lacked sufficient information to estimate latent constructs.

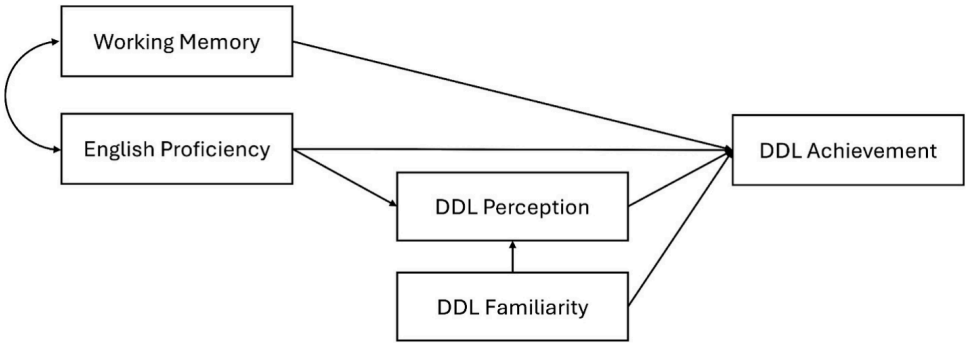



**Fig. 3.** Revised conceptual framework proposed for the meta-analytic path analysis.

#### *5.2.4. Coding moderator variables*

After finalizing the set of variables and their correlations for the path analysis model, we reviewed the remaining studies to code their methodological differences. This included, but was not limited to, factors such as target linguistic dimensions, learning environments, L1 backgrounds, age, and type of DDL instruction. In this study, moderator variables were coded from the selected studies only after confirming the path analysis model and the relationships between individual difference variables and learning outcomes. Typically, potential moderator variables are identified through theoretical review prior to data construction and analysis. However, in this study, we refrained from making assumptions about potential moderators in advance as no theoretical groundwork was considered meaningful for such conjectures. After coding the available moderators from the remaining studies, we identified the following five variables: (1) target linguistic domain (productive vs. receptive); (2) learning environment (classroom vs. remote learning); (3) L1-English proximity (related vs. unrelated); (4) DDL instruction type (hands-on vs. hands-off concordancing); and (5) learners' age (adolescent vs. adult). A detailed description of the coded moderator variables for the studies is provided in Appendix 3.

### 5.3. Data analysis step #1: pooled correlation matrix construction

Once all correlation coefficients were recorded, we synthesized them across all the variables from the studies into a pooled correlation matrix using the *metaSEM* package (Cheung & Chan, 2005, 2009) in R. This package enabled us to create the matrix by synthesizing the recorded correlation coefficients while assigning different weights based on sample sizes. Consequently, correlation coefficients from studies with higher precision (i.e., studies with larger samples) were assigned greater weight in constructing the pooled correlation matrix. In computing the pooled correlation coefficients, we further checked homogeneity of effect sizes to examine their influence on the correlation coefficients and generalizability. Also, we conducted an assessment for publication bias to assess the robustness of the findings.

### 5.4. Data analysis step #2: meta-analytic path analysis model construction

Using the meta-analytically constructed pooled correlation matrix, we built and validated a path analysis model, which is a specialized form of SEM. While SEM is typically employed for model validation or hypothesis verification, our study adopted a more exploratory design, prompted by the limited empirical research on the mechanisms underlying successful DDL in language learning. Our choice of path analysis was deliberate, as the model was constructed solely with observed variables and did not involve any latent constructs. Accordingly, we developed this path analysis model based on the identified correlations and variables, its pathways and relationships carefully drawn from existing literature to establish a theoretically sound framework. Model fit was assessed using five indices (Blunch, 2015): chi-square test statistics ($p > .05$), root mean square error of approximation (RMSEA <0.08), standardized root mean square residual (SRMR <0.08), comparative fit index (CFI > 0.90), and Tucker-Lewis index (TLI > 0.90).

## 6. Results

### 6.1. Pooled correlation matrix

As shown in Table 2, the computed correlation coefficients ranged from −0.001 to 0.21, and only two of them were statistically significant, such as correlations between (1) DDL achievement and (2) DDL familiarity ($r = 0.06$, SE = 0.03, $p = .047$) and between (1) DDL achievement and (4) English proficiency ($r = 0.21$, SE = 0.07, $p = .003$).

According to a checklist for such approaches (Kang & Ahn, 2021), it should be noted here that the overall test for homogeneity of effect sizes was highly significant ($Q = 521.73$, $df = 240$, $p < .001$), indicating substantial heterogeneity across the pooled correlation matrix. This global finding supports the decision to employ a follow-up MAPA approach for the meta-analysis of correlations. Furthermore, an assessment for publication bias using Egger's test indicated a statistically significant result ($b = −0.06$, $p = .004$), suggesting the presence of bias within the pooled correlation matrix. Nevertheless, the accompanying funnel plot (Fig. 4) revealed that the majority of the pooled correlation coefficients were symmetrically distributed within the plot, with the exception of a single outlying correlation.

**Table 2**
Pooled correlation matrix with five variables.

| No. | Variable | 1 | 2 | 3 | 4 | 5 |
|---|---|---|---|---|---|---|
| 1 | DDL Achievement | 1 | | | | |
| 2 | DDL Familiarity | 0.06* | 1 | | | |
| 3 | Perception of DDL | 0.02 | 0.01 | 1 | | |

| No. | Variable | 1 | 2 | 3 | 4 | 5 |
|---|---|---|---|---|---|---|
| 4 | English Proficiency | 0.21** | −0.001 | 0.02 | 1 | |
| 5 | Working Memory | 0.16 | 0.01 | 0.11 | 0.01 | 1 |

** $p < .01$.
* $p < .05$.

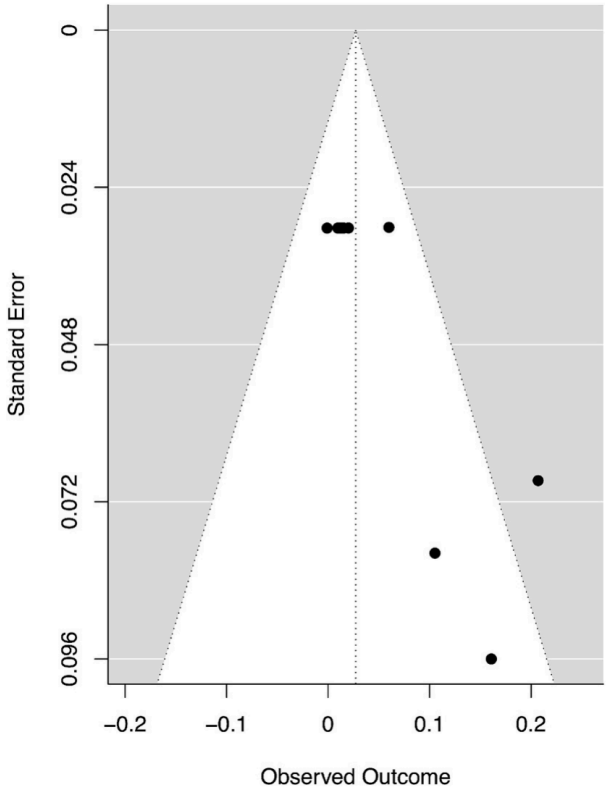



**Fig. 4.** Funnel plot for publication bias detection.

### 6.2. Meta-analytic path analysis model

The results of our MAPA approach indicated that the dataset fit the proposed model well, as demonstrated by five model fit indices: (1) $\chi^2(3) = 1.83$, $p = .61$; (2) CFI = 0.99; (3) TLI = 0.99; (4) RMSEA <0.001; and (5) SRMR = 0.03.

As shown in Table 3 and Fig. 5, analysis of the path coefficients revealed that English proficiency (β = 0.20, SE = 0.07, $p = .003$) and DDL familiarity (β = 0.06, SE = 0.03, $p = .046$) significantly influenced learning outcomes. The overall explanatory power of the model was modest ($R^2 = 0.071$), indicating that the four endogenous variables explained a small proportion (7.1%) of the variance in DDL achievement.

Although learners' positive perception of DDL played a central role as a mediator, it did not have a significant direct effect on DDL achievement. Consequently, no significant mediation effects were observed for English proficiency or DDL familiarity, indicating that these two

variables lacked significant indirect effects on DDL achievement. Furthermore, learners' working memory did not show a significant direct effect on DDL achievement. The mediating effects of the path analysis model are shown in Table 4.

Regarding the moderating effects, none of the identified five moderators, such as (1) target linguistic domain (productive vs. receptive), (2) learning environment (classroom vs. remote learning), (3) L1-English proximity (related vs. unrelated), (4) DDL instruction type (hands-on vs. hands-off concordancing), and (5) learners' age (adolescent vs. adult), showed any significant moderation effects. Specifically, (1) target linguistic domains, (3) L1-English proximity, and (4) DDL instruction type did not show significant moderation effects on any of the six path coefficients. Additionally, (2) learning environment and (5) learners' age lacked sufficient variability for further analysis as only one study (4%) was conducted outside the classroom, and only two studies (8%) included adolescents as participants.

**Table 3**
Estimates of path analysis model.

| Endogenous variables | Exploratory variables | Estimate | Standard errors | *p*-value |
|---|---|---|---|---|
| DDL achievement | Working memory | 0.16 | 0.10 | 0.10 |
| | English proficiency | 0.20 | 0.07 | 0.003 |
| | DDL perception | 0.01 | 0.03 | 0.76 |
| | DDL familiarity | 0.06 | 0.03 | 0.04 |
| DDL perception | English proficiency | 0.02 | 0.03 | 0.51 |
| | DDL familiarity | 0.01 | 0.03 | 0.72 |

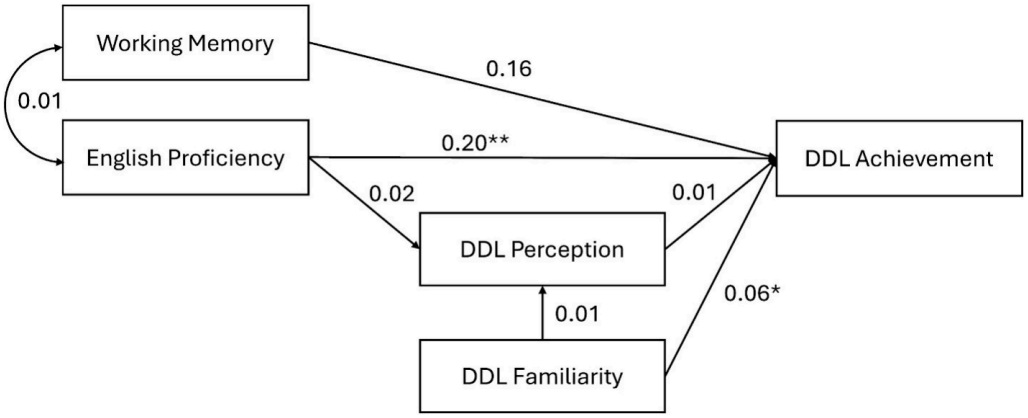



**Fig. 5.** Proposed path analysis model and their computed path coefficients.
*Note.* The model fit indices confirmed that the dataset successfully fitted the proposed path analysis model: (1) $\chi^2(3) = 1.83$, $p = .61$; (2) CFI = 0.99; (3) TLI = 0.99; (4) RMSEA <0.001; (5) SRMR = 0.03.

**Table 4**
Mediating effects of path analysis model.

| Endogenous variables | Exploratory variables | Direct effects | Indirect effects | Total effects |
|---|---|---|---|---|
| DDL achievement | Working memory | 0.16 | – | 0.16 |

| Endogenous variables | Exploratory variables | Direct effects | Indirect effects | Total effects |
|---|---|---|---|---|
| | English proficiency | 0.20** | 0.002 | 0.202** |
| | DDL perception | 0.01 | – | 0.01 |
| | DDL familiarity | 0.06* | 0.001 | 0.061* |
| DDL perception | English proficiency | 0.02 | – | 0.02 |
| | DDL familiarity | 0.01 | – | 0.01 |

## 7. Discussion

As one of the first synthetic efforts to examine correlation studies related to DDL, this study synthesized quantifiable correlation coefficients among individual difference variables and learning outcomes to investigate the mechanisms underlying successful DDL in language learning using a MAPA approach. This approach enabled us to examine complex interrelationships among DDL-related variables from a comprehensive perspective and offers several methodological advantages over analyzing multiple bivariate correlations in isolation.

Our analysis identified three cognitive factors—learners' working memory, English proficiency, and DDL familiarity—and one affective factor—learners' positive perception of DDL—as exploratory variables, with DDL achievement as the endogenous variable. The results revealed that learners' English proficiency and DDL familiarity had statistically significant direct impacts on DDL achievement, whereas learners' perception of DDL and working memory did not. Additionally, none of the exploratory variables exhibited indirect effects on DDL achievement. Although several moderator variables were identified, none showed statistically significant moderation effects on the path coefficients among the factors in the MAPA. The results of the present study offer valuable insights into the mechanisms of successful DDL in language learning, emphasizing the pivotal role of cognitive factors, limited influence of affective and memory-related variables, and stable nature of DDL outcomes across diverse contexts. These aspects are discussed in detail in the following subsections.

### 7.1. Cognitive factors as primary predictors of DDL success

The findings from this study highlight the critical role of cognitive factors—particularly learners' English proficiency and DDL familiarity—in achieving success in DDL-based language learning, corresponding to findings by Lee et al. (2020). English proficiency emerged as the strongest predictor of DDL achievement, consistent with SLA theories that emphasize the foundational role of linguistic competence in processing and applying new input (Boulton & Cobb, 2017; Lee et al., 2020). This finding supports previous literature, suggesting that learners' L2 proficiency determines whether they struggle to interpret complex linguistic structures and advanced vocabulary in DDL owing to limited proficiency, or whether

they can easily identify nuanced patterns and subtleties, in line with higher proficiency (e.g., Breyer, 2009; Flowerdew, 2012). It further indicates that learners with higher L2 proficiency can better recognize linguistic patterns and process information across the given concordance lines, so that they can navigate and interpret corpus data more efficiently than their peers with lower L2 proficiency.

DDL familiarity was also a significant predictor, highlighting the critical role of task-specific expertise in DDL success. This supports the idea that learners' perceptions and understanding of language learning tasks of how to handle the given tasks effectively are key factors in ensuring successful language learning (Gavioli, 2009), including DDL activities. As such, novice users of DDL may encounter initial challenges, such as the complexity of concordancing tools or unfamiliarity with data-analysis tasks. By contrast, more experienced learners can focus on interpreting linguistic patterns, rather than grappling with the mechanics of the tools themselves (Boulton & Cobb, 2017; Lee et al., 2020).

### 7.2. Limited role of affective and memory factors

Despite the theoretical relevance of learners' positive perception of DDL and working memory, neither variable demonstrated significant direct or indirect effects on DDL achievement. This finding highlights the importance of re-examining their roles in DDL contexts. The lack of impact from learners' perceptions suggests that although positive attitudes toward DDL can enhance engagement (Boulton & Cobb, 2017), they alone appear insufficient to produce measurable achievement gains—at least within the empirical evidence of our dataset. Motivation theories in SLA, such as Gardner's socio-educational model (Gardner, 1985, 2001)—a prominent framework in language learning motivation—highlight the critical role of attitudes in achieving language learning success. However, in the case of DDL, our findings suggest that the structured and analytical nature of tasks may prioritize procedural and linguistic skills over affective engagement. Similarly, the non-significance of working memory challenges its assumed necessity in DDL tasks, where analyzing complex linguistic data was thought to require significant cognitive resources (Lee et al., 2020). This finding suggests that corpus tools, such as concordancers, might offload or distribute cognitive demands, reducing reliance on internal capacities and facilitating learners' engagement with complex data.

### 7.3. Consistency of DDL outcomes across contexts

Although the finding related to no significant moderating effects contrasts with the expectations of previous studies (e.g., Boulton & Cobb, 2017), it suggests that the lack of significant moderation effects from contextual variables—such as target linguistic domains,

L1-English proximity, and DDL instruction type—points to the robust and consistent nature of the effectiveness of DDL. This finding suggests that the structural relationships observed in the current model could be broadly applicable across various learning contexts, including DDL for both productive (e.g., writing) and receptive (e.g., vocabulary acquisition and grammar knowledge) skills. These relationships hold for learners with different linguistic backgrounds, whether their L1s are related to English (e.g., French, Spanish, and Russian) or not (e.g., Chinese, Korean, and Thai). Additionally, the model applies to diverse DDL activities, ranging from hands-on concordancing (e.g., learner-led concordancing use) to hands-off approaches (e.g., paper-based DDL materials). However, this consistency may also reflect limitations in the current research landscape. The relatively few studies included in the meta-analysis means that moderator analyses—essentially subgroup analyses—may not have had sufficient statistical power to robustly verify the hypotheses regarding contextual influences (Jak & Cheung, 2018). Smaller sample sizes can lead to issues such as insufficient power to detect significant differences or effects, potentially leading to a failure to observe subtle but important moderators.

### 7.4. Pedagogical implications

This section outlines the pedagogical implications of the study. First, given the importance of L2 proficiency, instructors should consider incorporating materials tailored to learners' proficiency levels. For example, using simplified (or graded) corpora (e.g., Allan, 2009; Lee et al., 2015) and extracting input from them could be an effective approach. Alternatively, instructors can carefully select sentences from target corpora for their students (see Frankenberg-Garcia, 2012, 2014; Lee et al., 2017, for examples and discussion). Although this may be somewhat labor-intensive, recent developments in generative AI (GenAI) may assist teachers in this process (see below for further suggestions on integrating GenAI).

Second, the importance of DDL familiarity highlights the need for targeted scaffolding in DDL instruction. Educators should provide preparatory training to help learners become familiar with corpus tools. For instance, step-by-step guidance on how to perform concordance searches (e.g., Saeedakhtar et al., 2020; Vyatkina, 2016) can help reduce cognitive load for less experienced learners, thereby enabling them to engage more effectively with DDL tasks. Additionally, integrating DDL into GenAI-assisted environments (Crosthwaite & Baisa, 2023; Flowerdew, 2025; Lin, 2023)—platforms increasingly familiar to today's learners—may further improve DDL familiarity and contribute to more successful learning outcomes.

## 8. Conclusion and directions for future research

As the first study to apply a MAPA approach to DDL research, this investigation synthesized quantifiable correlations among individual difference variables and learning outcomes. This approach enabled us to examine the mechanisms underlying successful DDL in language learning. By identifying key predictors of DDL achievement and clarifying their structural relationships, the study offers valuable insights into how DDL can be effectively tailored to learners with diverse individual profiles.

Despite these contributions, the study has several limitations. First, the dataset included relatively few studies ($k$ = 25, $N$ = 1094), limiting the inclusion of other potentially influential variables. Second, while the exploratory meta-analytic model identified meaningful relationships, its exploratory nature and limited empirical foundation create a risk of overfitting, potentially limiting the model's generalizability. Third, certain potential moderators could not be examined due to insufficient variation in the data. For instance, only 4% of studies were conducted outside the classroom, and only 8% included adolescent learners, narrowing the applicability of our findings across age groups and settings.

In light of these limitations, future DDL research would benefit from broader and more diverse sampling strategies. Studies should aim to represent a wider range of educational settings, learner demographics, and delivery modes. For example, research involving adolescents could examine how developmental differences influence their interaction with DDL tools. Similarly, investigating DDL in online learning contexts, including GenAI-assisted ones (Flowerdew, 2025; Lin, 2023), could reveal how digital literacy and self-regulation affect outcomes, particularly compared to traditional classroom environments.

Additional research is also needed to unpack the role of affective and cognitive variables. While perception and memory may not directly predict DDL success, maintaining a positive attitude toward DDL and supporting working memory could be essential for sustained engagement. Future studies might explore whether affective factors such as motivation or willingness to adopt DDL tools play a greater role in long-term learning outcomes. Furthermore, investigating how task complexity interacts with cognitive load may help explain why working memory appears to have a limited direct effect on DDL performance.

In summary, this study underscores the importance of cognitive skills in DDL success and identifies several avenues for expanding our understanding of its mechanisms. Addressing these gaps through more comprehensive and targeted research can inform the design of more effective, equitable, and personalized DDL interventions, ultimately transforming language learning practices on a global scale.

## Educational relevance and implications statement

This study explores how individual differences, like memory, language skills, and experience with certain tools, affect how well people learn languages using data-driven learning (DDL). DDL involves using real-life examples from large collections of texts (called corpora) to discover patterns in language use. The research identifies key ingredients for successful DDL—such as learners' skills and familiarity with the approach—and reveals how these factors interact to influence learning outcomes. These findings deepen our understanding of the mechanisms behind DDL and can help educators design more personalized and effective teaching methods using real-world language data.

## CRediT authorship contribution statement

**Hansol Lee:** Writing – review & editing, Writing – original draft, Visualization, Resources, Methodology, Data curation, Conceptualization. **Jang Ho Lee:** Writing – review & editing, Writing – original draft, Validation, Supervision, Methodology, Investigation, Data curation, Conceptualization.

## Ethics statement

An ethics statement is not applicable because this study is based exclusively on published literature.


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## Declaration of competing interest

No potential conflict of interest was reported by the authors.


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# Appendix 1. List of 25 Studies

| No. | Study No. | Reference | Publication Type |
|---|---|---|---|
| 1 | 1 | Almazova, N., & Kogan, M. (2014). Computer assisted individual approach to acquiring foreign vocabulary of students major. In Learning and Collaboration Technologies. Technology-Rich Environments for Learning and Collaboration: First International Conference, LCT 2014, Held as Part of HCI International 2014, Heraklion, Crete, Greece, June 22–27, 2014, Proceedings, Part II 1 (pp. 248–257). Springer International Publishing. | Conference Presentation |
| 2 | 2 | Alex Boulton. (2012). Hands-on / hands-off: Alternative approaches to data-driven learning. In J. Thomas & A. Boulton (eds), Input, Process and Product: Developments in Teaching and Language Corpora. Brno: Masaryk University Press, p. 152–168. | Book Chapter |
| 3 | 4 | Chen, M., Flowerdew, J., & Anthony, L. (2019). Introducing in-service English language teachers to data-driven learning for academic writing. System, 87, 102148. | Journal Article |
| 4 | 5 | Curado-Fuentes, A. B. (2024) [1]. Digital reading in English for Business Management and Tourism: The importance of micro-skills for the curriculum. Research in Education and Learning Innovation Archives, (33), 16–34. | Journal Article |
| 5 | 6 | Curado-Fuentes, A. B. (2024) [2]. Digital reading in English for Business Management and Tourism: The importance of micro-skills for the curriculum. Research in Education and Learning Innovation Archives, (33), 16–34. | Journal Article |
| 6 | 7 | Dewiloh, K. (2022). The effects of data-driven learning (DDL) on the teaching of English phrasal verbs to Thai EFL secondary students (Doctoral dissertation, Thammasat University). | Dissertation |
| 7 | 8 | Đurović, Z., & Bauk, S. (2022). Testing corpus linguistics methods in ESP vocabulary teaching. Journal of Teaching English for Specific and Academic Purposes, 319–337. | Journal Article |
| 8 | 9 | Hirata, E. (2019). The development of a multimodal corpus tool for young EFL learners: A case study on the integration of DDL in teacher education. In P. Crosthwaite (Ed.), Data-Driven Learning for the Next Generation (pp. 88–105). Routledge. | Book Chapter |
| 9 | 11 | Kogan, M. S., Komarova, I. A., & Dmitrijev, A. V. (2020, July 16–17). Didactic potential of applying corpus approach to English phraseology studies and teaching [Conference presentation]. TaLC2020, Perpignan, France. | Conference Presentation |
| 10 | 12 | Kontsioti, V. (2024). Teaching collocations to teenagers through α fantasy-adventure movies/series subtitle corpus: A soft ddl approach (Doctoral dissertation, Aristotle University of Thessaloniki). | Dissertation |
| 11 | 13 | Lee, H., Warschauer, M., & Lee, J. H. (2020). Toward the establishment of a data-driven learning model: Role of learner factors in corpus-based second language vocabulary learning. The Modern Language Journal, 104(2), 345–362. | Journal Article |
| 12 | 15 | Lin, M. H. (2021). Effects of data-driven learning on college students of different grammar proficiencies: A preliminary empirical assessment in EFL classes. Sage Open, 11(3), 21582440211029936. | Journal Article |
| 13 | 16 | Liontas, J. I., Bangun, I. V., & Li, S. (2023). Lexical Collocational Instruction in EAP Writing via COCA. Teaching English With Technology, 23(1), 80–106. | Journal Article |
| 14 | 17 | Liou, H. C., & Liu, S. Y. (2021). Exploring the relationships between English writing motivation and uptake of corpus-aided corrective feedback: A longitudinal study. In Corpora in ESP/EAP Writing Instruction (pp. 57–77). Routledge. | Book Chapter |
| 15 | 18 | Liu, T., & Gablasova, D. (2025). Data-driven learning of collocations by Chinese learners of English: a longitudinal perspective. Computer Assisted Language Learning, 38(3), 612–637. https://doi.org/10.1080/09588221.2023.2214605 | Journal Article |
| 16 | 19 | Mizumoto, A., & Chujo, K. (2016). Who is data-driven learning for? Challenging the monolithic view of its relationship with learning styles. System, 61, 55–64. | Journal Article |
| 17 | 20 | Mizumoto, A., Chujo, K., & Yokota, K. (2016). Development of a scale to measure learners' perceived preferences and benefits of data-driven learning. ReCALL, 28(2), 227–246. | Journal Article |
| 18 | 21 | Quinn, C. (2018). Corpora as an L2 writing reference tool: Classroom practice and student response. In I. Walker, D. K. G. Chan, M. Nagami, & C. Bourguignon (Eds.), New perspectives on the development of communicative and related competence in foreign language education (pp. 309–338). De Gruyter Mouton. | Book Chapter |
| 19 | 23 | Todd, R. W. (2001). Induction from self-selected concordances and self-correction. System, 29(1), 91–102. | Journal Article |
| 20 | 24 | Vyatkina, N. (2016). Data-driven learning of collocations: Learner performance, proficiency, and perceptions. Language Learning & Technology, 20(3), 159–179. | Journal Article |
| 21 | 25 | Wen, F. C. (2016). Using corpora to aid in learning collocations (Doctoral dissertation, University of Illinois at Urbana-Champaign). | Dissertation |

| No. | Study No. | Reference | Publication Type |
|---|---|---|---|
| 22 | 27 | Yu, L., & Shen, J. (2022). Analysis of the correlation between academic performance and learning motivation in english course under a corpus-data-driven blended teaching model. Scientific Programming, 2022(1), 3407270. | Journal Article |
| 23 | 28 | Zare, J., & Aqajani Delavar, K. (2023). A data-driven learning focus on form approach to academic English lecture comprehension. Applied Linguistics, 44(3), 485–504. | Journal Article |
| 24 | 29 | Zare, J., & Karimpour, S. (2023). The Role of Working Memory in the Impact of Data-Driven Learning on Academic English Lecture Comprehension. Language Related Research. Advance Online Publication. | Journal Article |
| 25 | 30 | Zare, J., & Karimpour, S. (2022). Classroom Concordancing and Second Language Motivational Self-System: A Data-Driven Learning Approach. Frontiers in Psychology, 13, 841584. | Journal Article |

# Appendix 2. Dataset

| Study No. | *n* | Order | Variable | 1 | 2 | 3 | 4 | 5 |
|---|---|---|---|---|---|---|---|---|
| 1 | 6 | 01–1 | 1. DDL achievement | 1 | | | | |
| | | 01–2 | 2. DDL familiarity | | 1 | | | |
| | | 01–3 | 3. Perception of DDL | 0.76 | | 1 | | |
| | | 01–4 | 4. English proficiency | | | | 1 | |
| | | 01–5 | 5. Working memory | | | | | 1 |
| 2 | 108 | 02–1 | 1. DDL achievement | 1 | | | | |
| | | 02–2 | 2. DDL familiarity | 0.27 | 1 | | | |
| | | 02–3 | 3. Perception of DDL | | | 1 | | |
| | | 02–4 | 4. English proficiency | −0.32 | | | 1 | |
| | | 02–5 | 5. Working memory | | | | | 1 |
| 4 | 49 | 04–1 | 1. DDL achievement | 1 | | | | |
| | | 04–2 | 2. DDL familiarity | | 1 | | | |
| | | 04–3 | 3. Perception of DDL | | 0.25 | 1 | | |
| | | 04–4 | 4. English proficiency | | | | 1 | |
| | | 04–5 | 5. Working memory | | | | | 1 |
| 5 | 22 | 05–1 | 1. DDL achievement | 1 | | | | |
| | | 05–2 | 2. DDL familiarity | | 1 | | | |
| | | 05–3 | 3. Perception of DDL | | | 1 | | |
| | | 05–4 | 4. English proficiency | 0.45 | | | 1 | |
| | | 05–5 | 5. Working memory | | | | | 1 |
| 6 | 19 | 06–1 | 1. DDL achievement | 1 | | | | |
| | | 06–2 | 2. DDL familiarity | | 1 | | | |
| | | 06–3 | 3. Perception of DDL | 0.38 | | 1 | | |
| | | 06–4 | 4. English proficiency | 0.57 | | 0.04 | 1 | |
| | | 06–5 | 5. Working memory | | | | | 1 |
| 7 | 50 | 07–1 | 1. DDL achievement | 1 | | | | |
| | | 07–2 | 2. DDL familiarity | | 1 | | | |
| | | 07–3 | 3. Perception of DDL | | | 1 | | |
| | | 07–4 | 4. English proficiency | 0.27 | | | 1 | |
| | | 07–5 | 5. Working memory | | | | | 1 |
| 8 | 15 | 08–1 | 1. DDL achievement | 1 | | | | |
| | | 08–2 | 2. DDL familiarity | | 1 | | | |
| | | 08–3 | 3. Perception of DDL | | | 1 | | |
| | | 08–4 | 4. English proficiency | 0.914 | | | 1 | |
| | | 08–5 | 5. Working memory | | | | | 1 |
| 9 | 45 | 09–1 | 1. DDL achievement | 1 | | | | |
| | | 09–2 | 2. DDL familiarity | | 1 | | | |

| Study No. | *n* | Order | Variable | 1 | 2 | 3 | 4 | 5 |
|---|---|---|---|---|---|---|---|---|
| | | 09–3 | 3. Perception of DDL | | 0.03 | 1 | | |
| | | 09–4 | 4. English proficiency | | | | 1 | |
| | | 09–5 | 5. Working memory | | | | | 1 |
| 11 | 11 | 11–1 | 1. DDL achievement | 1 | | | | |
| | | 11–2 | 2. DDL familiarity | | 1 | | | |
| | | 11–3 | 3. Perception of DDL | | | 1 | | |
| | | 11–4 | 4. English proficiency | 0.59 | | | 1 | |
| | | 11–5 | 5. Working memory | | | | | 1 |
| 12 | 7 | 12–1 | 1. DDL achievement | 1 | | | | |
| | | 12–2 | 2. DDL familiarity | | 1 | | | |
| | | 12–3 | 3. Perception of DDL | | | 1 | | |
| | | 12–4 | 4. English proficiency | 0.36 | | | 1 | |
| | | 12–5 | 5. Working memory | | | | | 1 |
| 13 | 34 | 13–1 | 1. DDL achievement | 1 | | | | |
| | | 13–2 | 2. DDL familiarity | 0.3025 | 1 | | | |
| | | 13–3 | 3. Perception of DDL | | | 1 | | |
| | | 13–4 | 4. English proficiency | 0.36 | −0.03 | | 1 | |
| | | 13–5 | 5. Working memory | 0.4 | 0.31 | | 0.48 | 1 |
| 15 | 95 | 15–1 | 1. DDL achievement | 1 | | | | |
| | | 15–2 | 2. DDL familiarity | | 1 | | | |
| | | 15–3 | 3. Perception of DDL | | | 1 | | |
| | | 15–4 | 4. English proficiency | | | | 1 | |
| | | 15–5 | 5. Working memory | | | | | 1 |
| 16 | 16 | 16–1 | 1. DDL achievement | 1 | | | | |
| | | 16–2 | 2. DDL familiarity | | 1 | | | |
| | | 16–3 | 3. Perception of DDL | | | 1 | | |
| | | 16–4 | 4. English proficiency | 0.99 | | | 1 | |
| | | 16–5 | 5. Working memory | | | | | 1 |
| 17 | 16 | 17–1 | 1. DDL achievement | 1 | | | | |
| | | 17–2 | 2. DDL familiarity | | 1 | | | |
| | | 17–3 | 3. Perception of DDL | | | 1 | | |
| | | 17–4 | 4. English proficiency | | | | 1 | |
| | | 17–5 | 5. Working memory | 1.67 | | | | 1 |
| 18 | 34 | 18–1 | 1. DDL achievement | 1 | | | | |
| | | 18–2 | 2. DDL familiarity | | 1 | | | |
| | | 18–3 | 3. Perception of DDL | 0.143 | | 1 | | |
| | | 18–4 | 4. English proficiency | 0.79 | | −0.0015 | 1 | |
| | | 18–5 | 5. Working memory | | | | | 1 |
| 19 | 145 | 19–1 | 1. DDL achievement | 1 | | | | |
| | | 19–2 | 2. DDL familiarity | | 1 | | | |
| | | 19–3 | 3. Perception of DDL | | | 1 | | |
| | | 19–4 | 4. English proficiency | 0.7 | | | 1 | |
| | | 19–5 | 5. Working memory | | | | | 1 |
| 20 | 147 | 20–1 | 1. DDL achievement | 1 | | | | |
| | | 20–2 | 2. DDL familiarity | | 1 | | | |
| | | 20–3 | 3. Perception of DDL | | | 1 | | |
| | | 20–4 | 4. English proficiency | | | | 1 | |
| | | 20–5 | 5. Working memory | | | 0.6 | | 1 |
| 21 | 52 | 21–1 | 1. DDL achievement | 1 | | | | |
| | | 21–2 | 2. DDL familiarity | | 1 | | | |
| | | 21–3 | 3. Perception of DDL | | | 1 | | |

| Study No. | *n* | Order | Variable | 1 | 2 | 3 | 4 | 5 |
|---|---|---|---|---|---|---|---|---|
| | | 21–4 | 4. English proficiency | | | 0.32 | 1 | |
| | | 21–5 | 5. Working memory | | | | | 1 |
| 23 | 25 | 23–1 | 1. DDL achievement | 1 | | | | |
| | | 23–2 | 2. DDL familiarity | 0.76 | 1 | | | |
| | | 23–3 | 3. Perception of DDL | | | 1 | | |
| | | 23–4 | 4. English proficiency | | | | 1 | |
| | | 23–5 | 5. Working memory | | | | | 1 |
| 24 | 10 | 24–1 | 1. DDL achievement | 1 | | | | |
| | | 24–2 | 2. DDL familiarity | | 1 | | | |
| | | 24–3 | 3. Perception of DDL | | | 1 | | |
| | | 24–4 | 4. English proficiency | | | 0.46 | 1 | |
| | | 24–5 | 5. Working memory | | | | | 1 |
| 25 | 14 | 25–1 | 1. DDL achievement | 1 | | | | |
| | | 25–2 | 2. DDL familiarity | 0.51 | 1 | | | |
| | | 25–3 | 3. Perception of DDL | | | 1 | | |
| | | 25–4 | 4. English proficiency | | | | 1 | |
| | | 25–5 | 5. Working memory | | | | | 1 |
| 27 | 31 | 27–1 | 1. DDL achievement | 1 | | | | |
| | | 27–2 | 2. DDL familiarity | | 1 | | | |
| | | 27–3 | 3. Perception of DDL | | | 1 | | |
| | | 27–4 | 4. English proficiency | | | | 1 | |
| | | 27–5 | 5. Working memory | 1.88 | | | | 1 |
| 28 | 62 | 28–1 | 1. DDL achievement | 1 | | | | |
| | | 28–2 | 2. DDL familiarity | | 1 | | | |
| | | 28–3 | 3. Perception of DDL | | | 1 | | |
| | | 28–4 | 4. English proficiency | 0.01 | | | 1 | |
| | | 28–5 | 5. Working memory | | | | | 1 |
| 29 | 36 | 29–1 | 1. DDL achievement | 1 | | | | |
| | | 29–2 | 2. DDL familiarity | | 1 | | | |
| | | 29–3 | 3. Perception of DDL | | | 1 | | |
| | | 29–4 | 4. English proficiency | | | | 1 | |
| | | 29–5 | 5. Working memory | 0.21 | | | | 1 |
| 30 | 45 | 30–1 | 1. DDL achievement | 1 | | | | |
| | | 30–2 | 2. DDL familiarity | | 1 | | | |
| | | 30–3 | 3. Perception of DDL | | | 1 | | |
| | | 30–4 | 4. English proficiency | | | | 1 | |
| | | 30–5 | 5. Working memory | | | 1.74 | | 1 |

# Appendix 3. Moderator Variables

| Study No. | Receptive (1) Productive (0) | Classroom | L1-English Related (1) L1-English Not-related (0) | Hands-on Concordancing (1) Hands-off Concordancing (0) | Age |
|---|---|---|---|---|---|
| 1 | 1 | classroom | 1 | 1 | adult |
| 2 | 1 | classroom | 1 | 1 | adult |
| 4 | 0 | classroom | 0 | 1 | adult |
| 5 | 1 | classroom | 1 | 1 | adult |
| 6 | 1 | classroom | 1 | 1 | adult |
| 7 | 1 | classroom | 0 | 0 | adolescent |
| 8 | 1 | classroom | 1 | 0 | adult |
| 9 | 0 | classroom | 0 | 1 | adult |

| Study No. | Receptive (1) Productive (0) | Classroom | L1-English Related (1) L1-English Not-related (0) | Hands-on Concordancing (1) Hands-off Concordancing (0) | Age |
|---|---|---|---|---|---|
| 11 | 1 | classroom | 1 | 1 | adult |
| 12 | 1 | classroom | 1 | 0 | adolescent |
| 13 | 1 | classroom | 0 | 0 | adult |
| 15 | 1 | classroom | 0 | 0 | adult |
| 16 | 1 | classroom | 0 | 1 | adult |
| 17 | 0 | classroom | 0 | 1 | adult |
| 18 | 1 | classroom | 0 | 1 | adult |
| 19 | 0 | classroom | 0 | 1 | adult |
| 20 | 1 | classroom | 0 | 0 | adult |
| 21 | 0 | classroom | 0 | 1 | adult |
| 23 | 0 | classroom | 0 | 1 | adult |
| 24 | 0 | classroom | 1 | 1 | adult |
| 25 | 1 | classroom | 0 | 1 | adult |
| 27 | 1 | classroom | 0 | 1 | adult |
| 28 | 1 | at home | 0 | 0 | adult |
| 29 | 1 | classroom | 0 | 1 | adult |
| 30 | 1 | classroom | 0 | 1 | adult |

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